

FINAL LECTURE

What Now?

*From Optimization to
Continued Learning*

Linear Optimization and Network Flows

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ISYE

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FINAL LECTURE

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A Roadmap for What Comes Next

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Where We Have Been

A quick look back at the semester's arc

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Deeper LP Theory

Geometry, simplex, interior point, duality

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Sensitivity Analysis

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Books, courses, communities, papers

PART 01

Where We Have Been

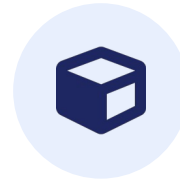
A semester's worth of optimization, in one glance.

What We Covered



Linear Programming

Formulating real-world problems as LPs; feasibility, optimality, interpretation of solutions.



Integer Programming

Binary decisions and logical constraints; branch-and-bound, cutting planes, solver behavior.



Network Flow Models

MCNF, max flow, shortest path, transportation, assignment, and matching — exploiting graph structure.



Large-Scale Methods

Column generation, Dantzig-Wolfe, Benders decomposition, and Lagrangian relaxation.

*A strong foundation.
But only the beginning.*

What now?

PART 02

Deepening LP Theory

Geometry · Simplex · Interior Point · Duality

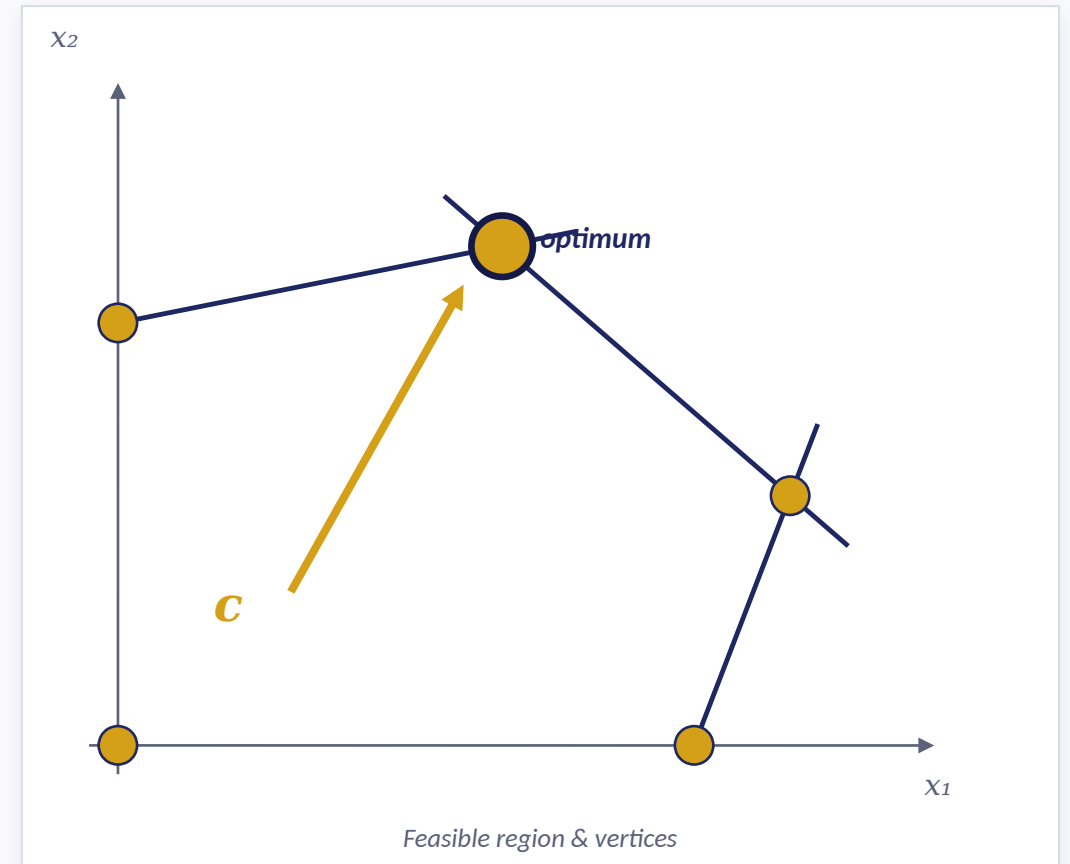
The Geometry of Linear Programming

Every LP has a geometric interpretation.

The feasible region of an LP is a **polyhedron** — the intersection of finitely many halfspaces. Optimal solutions, when they exist, occur at **vertices** (extreme points) of this polyhedron.

Key ideas to master:

- Convex sets and polyhedra
- Vertex / extreme ray representation (Minkowski-Weyl)
- Geometry of degeneracy, alternative optima, unboundedness



Read next: Bertsimas & Tsitsiklis, *Introduction to Linear Optimization* (Ch. 2–3) • Chvátal, *Linear Programming*

Two Algorithms, One Problem

THE SIMPLEX METHOD

Dantzig, 1947

Walks along the edges of the feasible polyhedron, pivoting from vertex to vertex.

Deepen your understanding:

- Revised simplex method
- Anti-cycling (Bland, lexicographic)
- Worst-case vs. practical complexity

INTERIOR POINT METHODS

Karmarkar, 1984

Moves through the interior of the feasible region — polynomial-time, competitive with simplex.

Deepen your understanding:

- Primal-dual path-following methods
- Central path & barrier functions
- Strong performance on large-scale LPs

Read next: Wright, *Primal-Dual Interior-Point Methods* • Nocedal & Wright, *Numerical Optimization* (Ch. 14)

Duality: A Deeper Look

Every LP has a dual LP.

The relationship between primal and dual is one of the most profound ideas in optimization — with consequences reaching into economics, game theory, and machine learning.

Strong duality

Optimal values coincide.

Complementary slackness

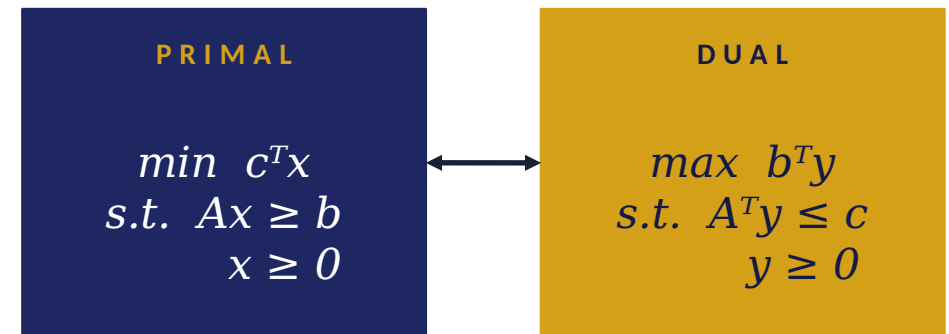
Links primal & dual solutions.

Shadow prices

Economic interpretation of dual variables.

Farkas' lemma

The theoretical bedrock.



At optimality

$$c^T x^* = b^T y^*$$

Strong duality theorem

PART 03

Sensitivity Analysis

Understanding the value of information.

A Question Every Modeler Must Answer

“How does the optimal solution change when the data change?”



Theory

Rooted in LP duality.

- Ranging analysis (objective, RHS)
- Shadow prices & reduced costs
- 100% rule for simultaneous changes
- Parametric programming



Practice

Solver-reported, instantly usable.

- Gurobi: dual values, reduced costs, basis status
- AMPL: .dual on constraints, .rc on variables
- Fast robustness checks on any optimal solution
- Essential skill — read your solver output

Separating a modeler who builds a model from one who draws actionable insight from it.

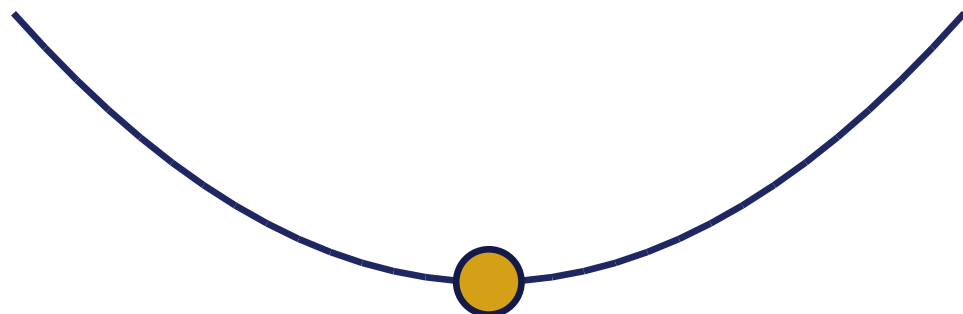
PART 04

Beyond Linear

Nonlinear · Convex · Non-convex Optimization

Convex vs. Non-Convex

CONVEX



global = local

Any local optimum is global.

QP · SOCP · SDP · LP → portfolios, SVMs, robust optimization

NON-CONVEX



Local optima $\neq$ global optimum.

Deep learning · design · combinatorial problems

THE book to read next: Boyd & Vandenberghe, **Convex Optimization** (free at web.stanford.edu/~boyd/cvxbook)

PART 05

Optimization Under Uncertainty

The real world doesn't hand you parameters — you estimate them.

Stochastic Programming & Robust Optimization



Stochastic Programming

Uncertainty as a probability distribution.

- Two-stage: decide now, recourse later
- Scenario-based formulations
- Natural for supply chains, energy, finance
- Requires distributional information

Read: Birge & Louveaux, *Introduction to Stochastic Programming*



Robust Optimization

Worst case over an uncertainty set.

- Protect against any realization in U
- No probability distribution needed
- Tractable for many problem classes
- Bertsimas & Sim: tunable conservatism

Read: Ben-Tal, El Ghaoui & Nemirovski, *Robust Optimization*

PART 06

Contemporary Applications

Optimization is a living methodology — deployed everywhere.

Where Your Skills Apply



Supply Chain & Logistics

Routing, facility location, inventory, distribution networks



Healthcare

OR scheduling, nurse staffing, organ matching, radiation therapy



Energy Systems

Unit commitment, dispatch, renewable integration, grid planning



Finance

Portfolio optimization, risk, algorithmic trading, credit scoring



Transportation

Airline crew & fleet, railway timetabling, ride-sharing



Manufacturing

Production scheduling, process optimization, quality control



Telecommunications

Network design, spectrum allocation, routing



Machine Learning

Regularized regression, SVMs, neural network training

Amazon, UPS, FedEx, United, the Cleveland Clinic, NYISO, Goldman Sachs — all rely on optimization at their core.

INFORMS Resources



INFORMS

The leading society for OR, analytics, and management science.

informs.org

“Saving lives, saving money, solving problems.”

— INFORMS tagline

Journals

Operations Research • Management Science • IJoC • IJoO • IJAA (Edelman Awards)

TutORials

Annual accessible intros to advanced topics — LLMs, ML, optimization

Annual Meeting

The largest gathering of OR professionals worldwide

Student Membership

Competitions, career resources, chapters

PART 07

AI, LLMs, and OR

Not a replacement. An amplifier.

What LLMs Can Do for the OR Workflow



Formulation

Draft mathematical models from natural-language problem descriptions — variables, objectives, constraints.



Code Generation

Produce solver code (Python/Gurobi, AMPL, PuLP, OR-Tools) from formulations, including data processing.



Debugging & Interpretation

Diagnose infeasibility, suggest relaxations, and explain solver output in plain language.



Learning & Literature

Summarize research papers, explain unfamiliar techniques, and navigate the OR literature.

You experienced this in this course — using Claude as a coding partner.

LLM-Powered Optimization Systems

OptiMUS

Ahmaditeshnizi, Gao, Udell (2024)

LLM agent that formulates and solves (MI)LP problems from natural language, with modular architecture.

OR-LLM-Agent

Zhang et al. (2025)

Reasoning-LLM framework (DeepSeek-R1) decomposing OR into modeling, code generation, and debugging.

ORLM

Tang et al. (2024)

Framework for training open-source LLMs for optimization modeling; introduces IndustryOR benchmark.

OPRO

Yang et al. (2023)

LLMs as optimizers — optimization task described in natural language, iteratively refined.

PaMOP

IJCAI 2025

Tree-structured decomposition guiding LLMs to generate AMPL code via partition-based prompting.

This field is moving fast. These systems are proof-of-concept today; tomorrow, they'll be standard tools.

What LLMs Cannot (Yet) Do



Mathematical rigor

Plausible-looking formulations can be subtly wrong — missing constraints, incorrect index sets, violated structure.



Algorithmic understanding

LLMs pattern-match from training data. Novel or unusual problem structures break them.



Scalability expertise

Decomposition choice, structure exploitation, solver tuning — these require expertise LLMs cannot reliably provide.



Domain context

Specific industries, regulations, stakeholder preferences — the things that shape real-world optimization.

The takeaway

Your optimization expertise becomes more valuable in the age of AI, not less.

PART 08

Working with Data

Models don't exist in a vacuum — they need data.

From Raw Data to Real Decisions



↑ where you live

SKILLS YOU'LL NEED

- Data wrangling (pandas, SQL)
- Statistical estimation (forecasting, cost modeling)
- Visualization (matplotlib, seaborn, Plotly)
- Connecting models to live data systems

RESEARCH FRONTIERS

- **Predict-then-optimize**
- Online optimization & learning (bandits, RL)
- Data-driven robust optimization
- *Elmachtoub & Grigas, "Smart Predict, then Optimize," MS 2022*

PART 09

Living, Studying, Competing with AI

This is not a future prediction. This is now.

Five Abilities That Set Professionals Apart

1

Formulate problems precisely.

AI generates code, but someone must define what problem to solve.

2

Evaluate AI output critically.

AI-generated solutions must be checked — theory lets you spot errors.

3

Combine domain knowledge with tools.

The most impactful professionals understand both method and context.

4

Learn continuously.

Tools, methods, best practices are evolving fast. Stay current.

5

Communicate results effectively.

Optimal solutions that can't be explained or trusted are worthless.

Optimization expertise is your competitive advantage.

PART 10

Resources for the Journey

Books, courses, communities, papers.

Core Textbooks for Next Steps

Topic	Book	Author(s)
LP theory & geometry	<i>Introduction to Linear Optimization</i>	Bertsimas & Tsitsiklis
LP (accessible intro)	<i>Linear Programming</i>	Chvátal
Convex optimization	<i>Convex Optimization</i>	Boyd & Vandenberghe (free online)
Nonlinear optimization	<i>Numerical Optimization</i>	Nocedal & Wright
Integer programming	<i>Integer Programming</i>	Conforti, Cornuéjols & Zambelli
Network flows	<i>Network Flows</i>	Ahuja, Magnanti & Orlin
Stochastic programming	<i>Introduction to Stochastic Programming</i>	Birge & Louveaux
Robust optimization	<i>Robust Optimization</i>	Ben-Tal, El Ghaoui & Nemirovski
Interior point methods	<i>Primal-Dual Interior-Point Methods</i>	Wright
Algorithms (modern)	<i>Algorithms for Optimization</i>	Kochenderfer & Wheeler

★ *If you read only one book next, make it Boyd & Vandenberghe.*

Courses, Solvers, and Communities



Online Courses

- Stephen Boyd's EE364a (Stanford, YouTube)
- MIT OCW 15.053 — Mathematical Programming
- Coursera / edX — Discrete Optimization



Solvers & Tools

- Gurobi — [gurobi.com](https://www.gurobi.com) (free academic)
- AMPL — ampl.com
- CVXPY — Python convex modeling
- Google OR-Tools, PuLP, Pyomo
- Open-source: HiGHS, SCIP, CBC, GLPK



Communities

- INFORMS — [informs.org](https://www.informs.org)
- OR Stack Exchange
- COIN-OR — coin-or.org
- OR/MS Today magazine

Most solvers offer free academic licenses — take advantage while you're a student.

Key Papers: AI Meets OR

- #1** Ahmaditeshnizi, Gao & Udell (2024). OptiMUS: Scalable Optimization Modeling with (MI)LP Solvers and LLMs. ICML.
- #2** Zhang et al. (2025). OR-LLM-Agent: Automating Modeling and Solving of OR Optimization Problems with Reasoning LLM. arXiv:2503.10009.
- #3** Tang et al. (2024). ORLM: A Customizable Framework in Training Large Models for Automated Optimization Modeling. arXiv:2405.17743.
- #4** Yang et al. (2023). Large Language Models as Optimizers. arXiv:2309.03409.
- #5** Elmachtoub & Grigas (2022). Smart "Predict, then Optimize." Management Science 68(1): 9–26.
- #6** Da Ros, Soprano, Di Gaspero & Roitero (2026). Large Language Models for Combinatorial Optimization: A Systematic Review. ACM Computing Surveys.

A PARTING THOUGHT

Optimization is not
just a course.

*It is a methodology,
a mindset,
and a career.*

Go use it.

Thank you.

It has been a privilege to teach you this semester.

Dr. Ziteng Wang

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